# Modern Reinforcement Learning: From Agents to LLMs

## Model-Free Prediction and Control

Lecture 2

Keywords: model-free learning, Monte Carlo, temporal-difference, bootstrapping, prediction, control, on-policy, off-policy, Q-learning, SARSA

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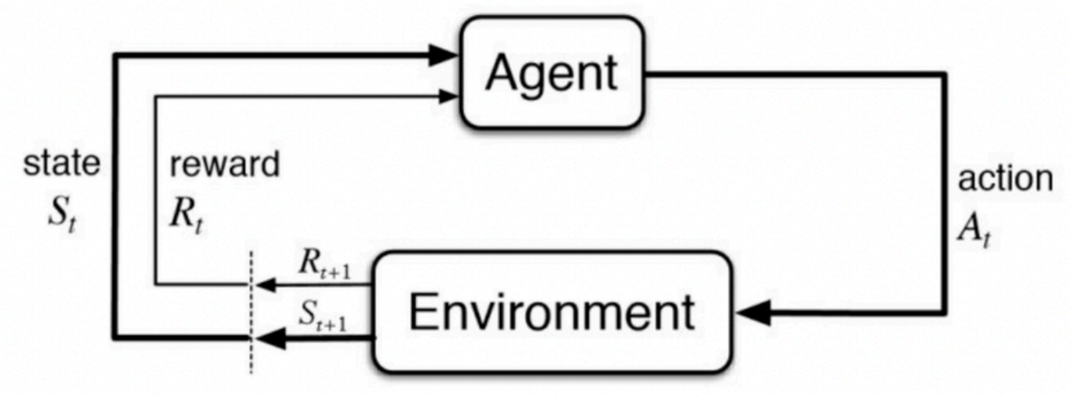
# 🎯 Goal

Understand how agents learn directly from experience without knowing the environment model

- We will:
  - Learn how to estimate value functions from sampled trajectories
  - Compare **Monte Carlo** (full-return) and **Temporal-Difference** (bootstrapped) methods
  - Extend prediction to **control** via **SARSA** and **Q-learning**

# From Dynamic Programming to Learning from Experience

- Lecture 1: **full knowledge** of transition dynamics $P(s' \mid s, a)$ and rewards $r(s, a)$
- But in most realistic settings, the model is **unknown** or **too complex**
- Agents must **learn directly from interaction** with the environment



# Why Model-Free Methods Matter

- Model-free RL ≈ **learn by doing** — estimate value from sampled experience
- No need for explicit environment equations
- Works in: games, robotics, reasoning, decision-making, dialogue systems
- Core challenge: balancing **exploration**, **variance**, and **efficiency**

$$\hat{V}(s) \approx \mathbb{E}\left[G_t \mid s_t = s\right]$$ estimated from sampled returns

# Experience as Data

- Each episode or trajectory provides samples of:

$$\tau^{\pi} = \left\{ (s_t, a_t, r_{t+1}, s_{t+1}) \right\}_{t=1}^{T}$$

- From these, we can approximate expectations → learn $V$ and $Q$
- Think of this as **data-driven value estimation**
- Learning signal comes from temporal structure of trajectories
- Rewards provide credit assignment to earlier states

# The Two Fundamental Approaches

| Approach | Core Idea | Pros | Cons |
|---|---|---|---|
| **Monte Carlo** | Learn from **complete returns** after each episode | Unbiased | High variance; episodic only |
| **Temporal-Difference (TD)** | Learn from **bootstrapped one-step targets** | Efficient, online | Biased estimates |

→ Two complementary paths to learn value from experience — both crucial for scalable reasoning and decision systems

# Monte Carlo: Learn from Complete Experience

$\tau_1$: $s_1 \to s_2 \to s_3 - s_4 \to s_{final}$

$\tau_2$: $s_1 \to s_2 \to s_{final}$

- Intuition: Learning by averaging returns
- Monte Carlo (MC) methods estimate value functions **from sampled returns**
- Learn by averaging observed outcomes over many episodes

Each episode → one full return $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$

# Monte Carlo: Learn from Complete Experience

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**Idea:** the vlaue of a state is its **average return** over many visits

$$V(s) \approx \frac{1}{N(s)} \sum_{i=1}^{N(s)} G_t^{(i)} \quad \text{where } s_t^{(i)} = s$$

# Estimating State Values from Sampled Returns

- **Goal:** estimate value function $V(s) = \mathbb{E}[G_t \mid s_t = s]$
- We do this by **averaging observed returns** for each state:
  - Each time state $s$ is visited, record the return $G_t$
  - Average all returns observed for $s$
- This gives an **unbiased estimate** of $V(s)$

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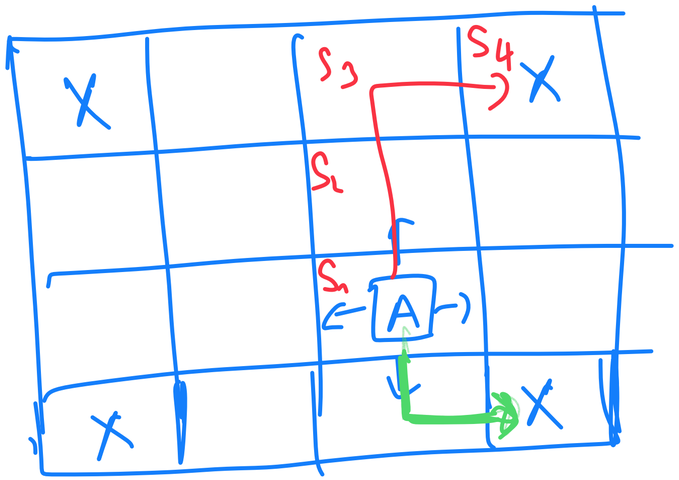
  - Average all returns observed for $s$
- This gives an **unbiased estimate** of $V(s)$
- Works only for **episodic tasks** where full $G_t$ can be computed

$$V(s) \leftarrow V(s) + \alpha\left[G_t - V(s)\right]$$ (incremental mean update form)

# Monte Carlo Prediction Example: Learning from Full Episodes




$r = -1 \;,\; 0 \;;\; \gamma = 0.9$

$s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_4$

$\quad\quad -1 \quad\; -1 \quad\; 0$

$$G_t = -1 + 0.9 \cdot (-1) + 0.9^2 \cdot (0)$$

$$= -1.9$$

# What Happens After One Episode

- Episode: $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_4$(terminal)
- After the episode finishes, you compute returns: $G_1$ for $s_1$, $G_2$ for $s_2$ , $G_3$ for $s_3$
- Then:  Update each visited state using its own return

$$V(s_1) \leftarrow V(s_1) + \alpha(G_1 - V(s_1))$$

$$V(s_2) \leftarrow V(s_2) + \alpha(G_2 - V(s_2))$$

$$V(s_3) \leftarrow V(s_3) + \alpha(G_3 - V(s_3))$$

- Terminal typically not updated (value fixed = 0)

# Two Ways to Average Returns: First-Visit vs. Every-Visit MC

- **First-Visit MC**
  - Use only the first occurrence of each state within an episode
  - Each state is updated at most once per episode
  - Update: $V(s) \leftarrow V(s) + \alpha\left[G_t - V(s)\right]$ where t is the first time $s_t = s$
- **Every-Visit MC**
  - Use all occurrences of a state within an episode
  - Update after every visit: $V(s) \leftarrow V(s) + \alpha\left[G_t - V(s)\right]$

# Convergence by the Law of Large Numbers

Monte Carlo estimate: $\hat{V}(s) \approx \frac{1}{N(s)} \sum_{i=1}^{N(s)} G_i(s)$

Law of Large Numbers (LLN): $\lim_{N(s) \to \infty} \hat{V}(s) = V^{\pi}(s)$

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- Interpretation
  - Each episode provides a noisy sample of the true value
  - Averaging across many episodes reduces noise while preserving the expectation
- Properties
  - Unbiased: expected estimate equals the true value
  - High variance: stochastic returns lead to slow convergence
  - Episodic only: requires full return $G_t$ cannot update before the episode ends

# From Prediction to Control: Monte Carlo with Exploring Starts

**Goal:** Learn optimal policy $\pi^*$ using Monte Carlo estimates of action values

**Key idea:**

- Use **returns from episodes** to estimate $Q(s, a)$

- Apply **Generalized Policy Iteration (GPI)**:

① policy eval. :

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[ G_t - Q(s,a) \right]$$

② policy iter. :

$$\pi(s) \leftarrow \arg\max_a Q(s,a)$$

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**Exploring starts assumption:**

- Each episode starts with a random $(s, a)$ pair

- Ensures all actions are tried infinitely often

**Consequence:**

- Monte Carlo control converges to optimal $Q^*$ and $\pi^*$

- In practice, exploring starts are hard to guarantee $\rightarrow$ leads to $\varepsilon$-greedy exploration (next slide).

# Balancing Exploration and Exploitation

- Motivation: Exploring starts are impractical — we need exploration within episodes

- ε-greedy policy $\pi(a \mid s)$:

$$a = \begin{cases} \arg\max_a Q(s, a) & \text{with prob. } 1 - \varepsilon \\ \text{random action} & \text{with prob. } \varepsilon \end{cases}$$

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**Algorithm (on-policy MC control):** Repeat:

1) Generate episodes using current ε-greedy $\pi$

2) Evaluate $Q(s,a)$ by averaging sample returns

3) Improve policy → make it greedy w.r.t. updated $Q$

**Guarantee:** With sufficient exploration $\varepsilon > 0$ and GLIE (greedy in the limit with infinite exploration),

on-policy MC control → $Q^*$ and $\pi^*$

# Off-Policy Monte Carlo Control (Importance Sampling)

- **Idea:** Learn the target policy $\pi$ using episodes generated by a different behavior policy $b$ ($\pi \neq b$)

- **Goal:** Estimate $V^{\pi}(s) = \mathbb{E}_{\pi}\left[G_t \mid S_t = s\right]$ using data collected from $b$

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- Importance Sampling (IS):
  - Correct returns via likelihood ratio:

$$G_t^{\text{corr}} = \rho_t G_t, \qquad \rho_t = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}$$

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- Corrects mismatch between behavior and target policies
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- Estimators:

Ordinary IS: $V^\pi(s) = \frac{1}{N} \sum_{i=1}^{N} \rho_t^{(i)} G_t^{(i)}$

Weighted IS (preferred): $V(s) \leftarrow \frac{\sum_i \rho_i G_i(s)}{\sum_i \rho_i}$ (normalize weights → lower variance)

**Interpretation:**
- Corrects mismatch between behavior and target policies
- Enables reuse of experience from any data-collecting policy

# Summary: Monte Carlo Learning & What Comes Next

- Core idea: Learn $V^\pi(s)$ or $Q^\pi(s, a)$ from sampled episodes. Update after the episode ends:

$$V(s) \leftarrow V(s) + \alpha\,[G_t - V(s)]$$

- Variants:
  - First-Visit vs Every-Visit
  - On-Policy (ε-greedy) vs Off-Policy (Importance Sampling)
- Strengths:
  - Unbiased estimates (directly from data)
  - Simple to implement, no model required
- Weaknesses:
  - High variance → slow convergence
  - Naturally suited to episodic settings only

📘 *Temporal-Difference (TD)* updates **after every step**, combining sampling with **bootstrapping.**

# Temporal-difference (TD) learning

# Learning from Incomplete Episodes: Temporal-Difference Methods

- Motivation
  - Monte Carlo (MC): learns from complete returns $G_t$
  - Dynamic Programming (DP): bootstraps using the true value $V(s')$
  - Temporal-Difference (TD): bootstraps using the current estimate $\hat{V}(s')$

- TD(0) Update Rule (updates afer every transition):

$$V(S_t) \leftarrow V(S_t) + \alpha\left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\right]$$

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- TD Error (Bellman Residual)

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

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- Incremental form:

$$V(S_t) \leftarrow V(S_t) + \alpha\delta_t$$

# Learning from Incomplete Episodes: Temporal-Difference Methods

Monte Carlo Target (Full Return): $$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots + \gamma^{T-t-1} R_T$$

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Key Differences:

- Monte Carlo (MC) uses the complete future return
- Temporal-Difference (TD) uses a bootstrapped estimate of the future return
- TD learns online, while MC must wait until the episode ends
- TD introduces bias but reduces variance → faster learning in practice

# 🧮 TD(0) Prediction Algorithm (for a given policy π)

‹/› Code

```
Input:  policy π, step-size α, discount γ
Initialize V(s) arbitrarily for all states s

Loop over episodes:
    Initialize S
    Repeat (for each step of episode):
        Take action A ~ π(· | S)
        Observe reward R and next state S'
        δ ← R + γ V(S') − V(S)      (TD error)
        V(S) ← V(S) + α δ           (TD update)
        S ← S'
    until S is terminal
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Notes:

- Online & incremental: updates after every transition

- Model-free: does not require transition probabilities

- Convergence: in the tabular setting with fixed policy $\pi$, converges to $V^\pi$ under standard step-size conditions

# TD as Stochastic Approximation to the Bellman Operator

- Bellman Expectation Equation

$$V^{\pi}(s) = \mathbb{E}_{\pi}\left[R_{t+1} + \gamma V^{\pi}(S_{t+1}) \mid S_t = s\right]$$

- Bellman Operator

$$(T^{\pi}V)(s) = \mathbb{E}_{\pi}\left[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s\right]$$

- Fixed-Point Property

$$V^{\pi} = T^{\pi}V^{\pi}$$

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---

- TD(0) as Stochastic Approximation

$$V(S_t) \leftarrow V(S_t) + \alpha\Big(R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\Big)$$

- The term inside parentheses is a sample of

$$(T^\pi V)(S_t) - V(S_t)$$

- Key Interpretation: TD performs noisy stochastic-gradient-like steps toward the fixed point of $T^\pi$

# 📉 Why TD Converges: $T^\pi$ is a $\gamma$-Contraction

- γ-Contraction Property: For any value functions

$$\|T^\pi V - T^\pi U\|_\infty \leq \gamma \|V - U\|_\infty \text{ with } \quad 0 \leq \gamma < 1$$

- Consequences
  - Unique fixed point: $V^\pi = T^\pi V^\pi$
  - Iterative convergence: $V_{k+1} = T^\pi V_k \;\Rightarrow\; V_k \to V^\pi$

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- TD(0): stochastic approximation of this contraction → convergence (under standard conditions)

- Key Takeaway: **contraction ⇒ unique fixed point ⇒ convergence**

# Why TD(0) Converges: Stochastic Approximation View

- TD(0) Update: $V(S_t) \leftarrow V(S_t) + \alpha_t\big(R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\big)$
- Rewrite as Noisy Fixed-Point Iteration

$$V(S_t) \leftarrow V(S_t) + \alpha_t\big((T^\pi V)(S_t) - V(S_t) + \text{noise}_t\big)$$

- where

$$\text{noise}_t = \underbrace{R_{t+1} + \gamma V(S_{t+1})}_{\text{sample}} - \underbrace{\mathbb{E}_\pi[R_{t+1} + \gamma V(S_{t+1}) \mid S_t]}_{(T^\pi V)(S_t)} \quad \text{and} \quad \mathbb{E}[\text{noise}_t \mid S_t] = 0.$$

- Convergence (Tabular Case)

$\sum_t \alpha_t = \infty, \qquad \sum_t \alpha_t^2 < \infty,$ and all states are visited infinitely often, $V_t \to V^\pi$ with probability 1

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# Example: How TD Propagates Value Backwards Step-by-Step

## Example: How TD Propagates Value Backwards Step-by-Step

*The +1 signal ripples back one cell per episode — watch it propagate.*

KEY INSIGHT The +1 reward signal is **propagated backward one step at a time**. Each episode, TD pushes value one cell further from the terminal — learning *incrementally* without waiting for episodes to finish.

4 × 4 GRIDWORLD

Ready

| | | | |
|---|---|---|---|
| 0,0<br>0<br>V(0,0) | 0,1<br>0<br>V(0,1) | 0,2 ★<br>–<br>V(0,2) | 0,3<br>0<br>V(0,3) |
| 1,0<br>0<br>V(1,0) | 1,1<br>0<br>V(1,1) | 1,2<br>0<br>V(1,2) | 1,3<br>0<br>V(1,3) |
| 2,0<br>0<br>V(2,0) | 2,1<br>0<br>V(2,1) | 2,2<br>0<br>V(2,2) | 2,3<br>0<br>V(2,3) |
| 3,0<br>0<br>V(3,0) | 3,1<br>0<br>V(3,1) | 3,2<br>0<br>V(3,2) | 3,3<br>0<br>V(3,3) |

Terminal (+1) Agent path Updating now Value changed

SCENARIO

Terminal = (0,2) ★, reward +1

Other transitions: R = 0 | α = 0.1 | γ = 1

Same path each episode: (2,3)→(2,2)→(1,2)→(0,2)

EPISODE 1 — TD(0) UPDATES

TRANSITION 1
(2,3) → (2,2), R = 0

TRANSITION 2
(2,2) → (1,2), R = 0

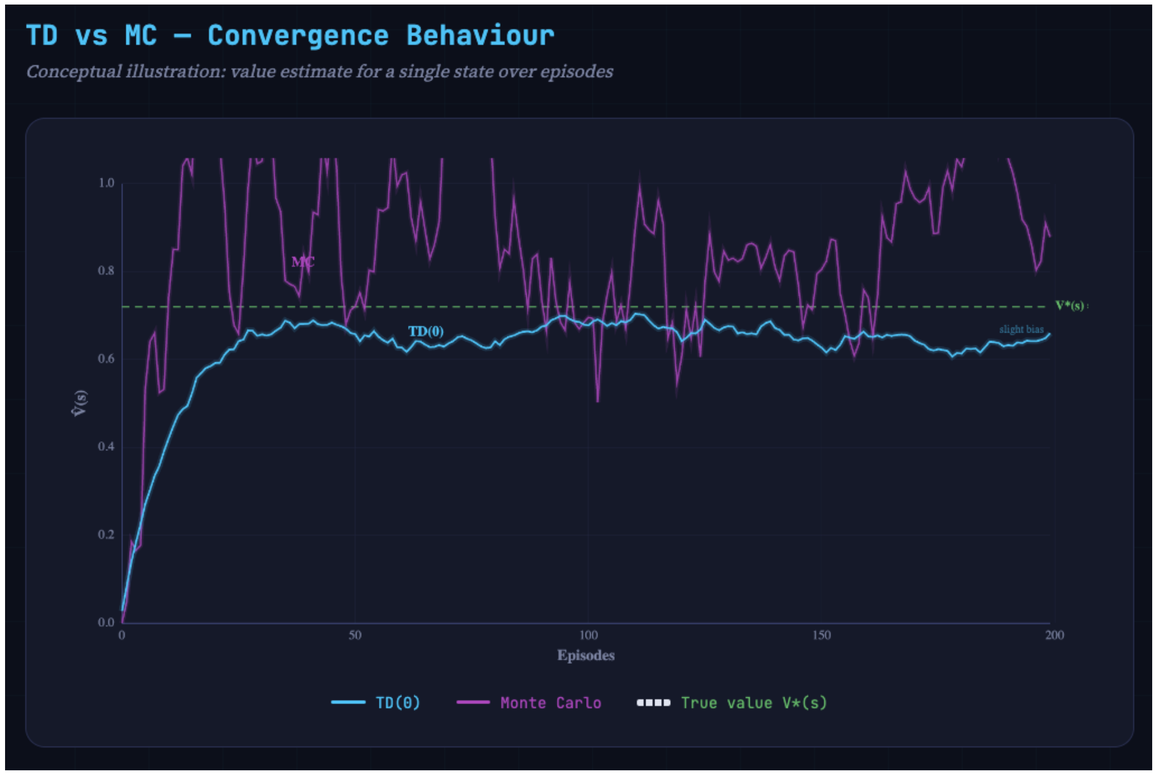
TRANSITION 3
(1,2) → (0,2), R = +1

← Back | Next → | Reset

Ready — press Next

# ⚖️ TD Learns Faster, but TD Estimates Can Be Biased

- TD(0) Converges Faster
  - Updates happen online
  - Lower variance → smoother learning
  - Typically reaches good performance sooner
- ⚠️ But TD Can Be Biased (Early On)
  - Bias comes from bootstrapping TD target
  - If $V(S_{t+1})$ is wrong, the error propagates backward




- Monte Carlo
  - Produces unbiased estimates
  - But: Must wait for full episodes; High variance → noisy learning curves

# 📊 Monte Carlo vs. Temporal-Difference (TD)

| Aspect | Monte Carlo (MC) | Temporal-Difference (TD) |
|---|---|---|
| **Learning target** | Full return $G_t$ | One-step bootstrap $R_{t+1} + \gamma V(S_{t+1})$ |
| **Update timing** | End of episode only | Every time-step |
| **Requires episodes?** | Yes (episodic only) | No (works in continuing tasks) |
| **Bias** | Unbiased | Biased (uses estimated future value) |
| **Variance** | High variance | Low variance |
| **Data efficiency** | Inefficient (delayed updates) | Efficient (online, incremental) |
| **Dependence on model** | Model-free | Model-free |
| **Convergence** | Slow | Faster, more stable |

# Model-free control

# Model-Free Control

- What is Control in RL?
- **Goal:** learn an optimal policy $\pi^*$ without knowing the model
- Only interact with the environment: observe (s, a, r, s')
- Learn optimal action–value function Q*(s, a)
- Derive policy by acting greedily with respect to Q*(s, a)

# On-Policy vs Off-Policy Control

- **Two fundamental approaches:**
  - On-policy control: learn the value of the policy that is actually used to generate behavior
  - Off-policy control: learn the value of a different (usually greedy) target policy while following a more exploratory behavior policy
  - Key distinction lies in the target used during value updates

- Definitions:
  - Behavior policy: $\pi_b(a \mid s)$
  - Target policy: $\pi_t(a \mid s)$

on-policy : $\pi_t = \pi_b$

off-policy : $\pi_t \neq \pi$

# SARSA: On-Policy TD Control (Intuition)

- **Key idea:**
  - Learn Q(s, a) while following the same policy that is being evaluated
  - Update target uses the action actually taken:

$$\text{Target} = r + \gamma Q(s', a')$$

- **Why SARSA?**
  - Naturally incorporates exploratory actions
  - Tracks the true return of the behavior policy
  - Safer in stochastic environments

- **Name:** S → A → R → S → A (state, action, reward, next state, next action)

# SARSA Update Rule (On-Policy)

- TD target (on-policy): $\text{Target} = r(s,a) + \gamma Q(s', a')$

- SARSA update:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[ \text{Target} - Q(s,a) \right]$$

- Characteristics:
  - On-policy: depends on the action $a'$ selected by the current behavior policy
  - Learns $Q^\pi$ where $\pi$ is typically ε-greedy w.r.t. $Q$
  - Converges under GLIE conditions

# SARSA Pseudocode (On-Policy TD Control)

- **SARSA (for action-value control):**

```
Initialize Q(s, a) arbitrarily

For each episode:
    Initialize state s
    Choose action a ~ ε-greedy(Q)

    Repeat:
        Take action a
        Observe reward r and next state s'
        Choose next action a' ~ ε-greedy(Q)

        Q(s, a) ← Q(s, a) + α [ r + γ Q(s', a') − Q(s, a) ]

        s ← s'
        a ← a'
    until s is terminal
```

# 2 Q-Learning: Off-Policy TD Control (Intuition)

- **Key idea:**
  - Learn the value of the greedy policy, even while following an exploratory behavior policy
  - Update target uses the best possible next action, not the action actually taken

- **Target:**

$$r + \gamma \max_{a'} Q(s', a')$$

- **Why Q-learning?**
  - Off-policy: behaves with ε-greedy policy, but learns about the greedy policy
  - Converges to optimal Q* under standard assumptions
  - Often learns faster toward optimal behavior

# Q-Learning Update Rule (Off-Policy)

- **TD target (off-policy):**

$$\text{Target} = r + \gamma \max_{a'} Q(s', a')$$

- **Q-learning update:**

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[ \text{Target} - Q(s,a) \right]$$

- Characteristics:
  - Off-policy: update uses greedy action, not the action actually taken
  - Learns the optimal action-value function Q*
  - Behavior policy can still be ε-greedy for exploration

# Q-Learning Pseudocode (Off-Policy TD Control)

- **Q-learning algorithm:**

```
Initialize Q(s, a) arbitrarily

For each episode:
    Initialize state s

    Repeat:
        Choose action a ~ ε-greedy(Q)
        Take action a
        Observe reward r and next state s'

        Q(s, a) ← Q(s, a) + α [ r + γ max_{a'} Q(s', a') - Q(s, a) ]

        s ← s'
    until s is terminal
```

# SARSA vs. Q-Learning: Summary

- SARSA (on-policy):
  - Target uses the action actually taken: Target = $r + \gamma Q(s', a')$
  - Learns $Q^{\pi}$ for the ε-greedy behavior policy
  - Accounts for exploratory actions

# SARSA vs. Q-Learning: Summary

- SARSA (on-policy):
  - Target uses the action actually taken: Target = $r + \gamma Q(s', a')$
  - Learns $Q^\pi$ for the ε-greedy behavior policy
  - Accounts for exploratory actions

- Q-Learning (off-policy):
  - Target uses the greedy action: Target = $r + \gamma \max_{a'} Q(s', a')$
  - Learns optimal $Q^*$ even while exploring
  - More aggressive; can converge faster

- Core difference:

SARSA evaluates the behavior policy.
Q-learning evaluates the greedy target policy.

# Extensions of Q-Learning

**Double Q-Learning**

- Addresses Q-learning's overestimation bias
- Maintain two value tables $Q_1$ and $Q_2$
- Update with cross-estimation: Use $Q_1$ to select argmax, $Q_2$ to evaluate it (or vice versa)
- More stable and accurate in stochastic environments

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**Dyna-Q**

- Integrates model learning + planning + real experience
- Learn a model from transitions $(s, a) \rightarrow (r, s')$
- Perform additional "imagined" Q-learning updates from the model
- Improves sample efficiency dramatically

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**Replay Buffers (Experience Replay)**

- Store past transitions in a buffer
- Sample mini-batches for off-policy Q-learning updates
- Breaks correlation between sequential samples
- Enables stable deep Q-learning (DQN)

# Big Picture Summary

- **Monte Carlo (MC) vs Temporal-Difference (TD)**
  - MC: full returns, unbiased but high variance
  - TD: sampled + bootstrapped target, biased but low variance
  - TD updates online; MC needs complete episodes
- **Prediction vs Control**
  - Prediction: learn $V^\pi$ or $Q^\pi$ for a fixed policy
  - Control: improve policy toward $\pi^*$ using SARSA or Q-learning
  - Control requires exploration (ε-greedy)
- **Core insight**
  - TD enables scalable, online RL and forms foundation for modern algorithms

# Why This Matters for LLMs

- Learning from sampled feedback
  - LLMs learn from sampled completions and reward model or preference scores
  - Bootstrapped reasoning
  - Reward model predictions serve as “value estimates”
- Updates propagate feedback backward through tokens
- Advantage estimation foundations
  - TD structure underpins advantage estimation (GAE) used in PPO/GRPO
  - Modern LLM RL fine-tuning builds directly on TD concepts

# Historical Note

- **Early milestones**
  - TD learning introduced in the 1980s (Sutton)
  - SARSA, Q-learning (Watkins) define classic RL control
  - Enabled first successful RL systems in robotics, games
- **Modern impact**
  - Foundations for deep RL (DQN, Atari breakthrough in 2015)
  - Basis for PPO, GRPO, A2C/A3C advantage-based algorithms
  - Still central in modern agentic LLM training
- These “classical” algorithms remain deeply embedded in today’s large models

# Reading Guide

- **Primary text:**
  - Sutton & Barto, Reinforcement Learning: An Introduction
    - Chapter 5 — Monte Carlo methods
    - Chapter 6 — Temporal-Difference learning

- **Optional deeper resources:**
  - Watkins (1989), Q-learning
  - Sutton (1988), Learning to predict by the methods of temporal differences
  - van Hasselt (2010), Double Q-learning
  - Dyna-Q and early model-based RL papers (Sutton 1990)

- Recommendation: Focus on the Bellman operator view and bias–variance intuition